A Minimal Research Project Template

An Illustrative Example in Python

Course Template

Your University

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Motivation

About This Template

- A minimal reproducible research workflow in Python
- Data flow: **pull** $\rightarrow$ **prepare** $\rightarrow$ **analyze** $\rightarrow$ **present**
- Built on the `mtcars` dataset as a teaching example
- Each stage writes a durable artifact to disk; the paper and presentation read from the same results file

The Research Question

- How does vehicle weight relate to fuel efficiency?
- Does transmission type matter after controlling for weight?
- Sample: 32 vehicles from the mtcars dataset
- Average fuel efficiency: 20.1 mpg; heaviest vehicle: 15

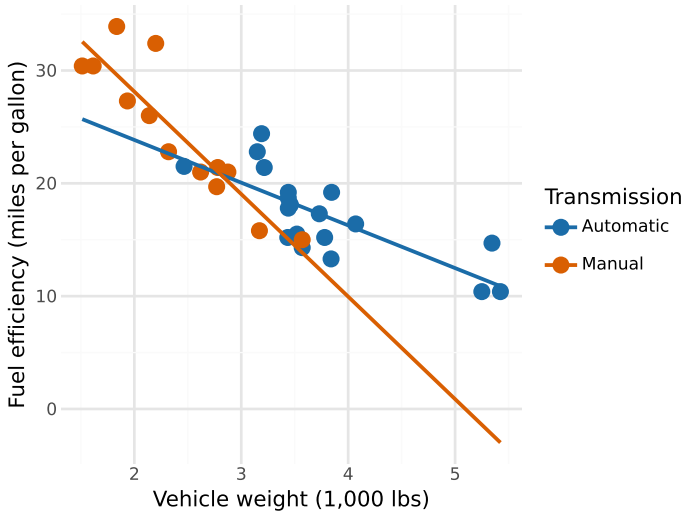
Data

Descriptive Statistics

Transmission	Cars	Mean fuel efficiency (mpg)	Mean horsepower	Mean weight (1,000 lbs)
Automatic	19	17.1	160.3	3.8
Manual	13	24.4	126.8	2.4

Results

Weight vs. Fuel Efficiency



Key Findings

- A 1,000 lb increase in weight is associated with a -**5.35 mpg** change
- Manual transmission cars show a -**0.02 mpg** difference vs. automatic after controlling for weight
- The prepared sample suggests a clear negative relationship between vehicle weight and fuel efficiency, while the manual cars in this small dataset have somewhat higher fuel efficiency after controlling for weight.

Conclusion

Takeaways

- The template keeps all moving parts visible while staying small enough to understand in one sitting
- Replace the toy `mtcars` data with your own; the workflow shape stays the same
- The paper and this presentation both read from `output/results.pkl` — run the pipeline once, render both outputs